

Q1. As shown in Fig. 1, NMOS transistor M_1 is part of a CS amplifier, and it operates in inversion region.

(1) Derive the expression of transistor intrinsic gain A_v , defined as $g_m r_o$, in terms of bias current I_D and the size W/L of M_1 .

(2) Explain why A_v cannot be increased by simply increasing I_D .

Solution:

$$(a) \quad I_D = \frac{1}{2} k'_n \frac{W}{L} (V_{GS} - V_{TH})^2 (1 + \lambda V_{DS}) \rightarrow V_{GS} - V_{TH} = \sqrt{\frac{2I_D}{k'_n \frac{W}{L} (1 + \lambda V_{DS})}}$$

$$g_m = \frac{\partial I_D}{\partial V_{GS}} = k'_n \frac{W}{L} (V_{GS} - V_{TH}) (1 + \lambda V_{DS})$$

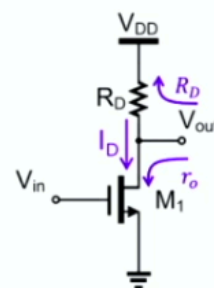


Fig. 1 CS amplifier with resistive load

Replace $V_{GS} - V_{TH}$ in g_m with its expression in I_D and V_{DS}

$$g_m = \sqrt{2k'_n \frac{W}{L} I_D (1 + \lambda V_{DS})}$$

$$r_o = \frac{\partial V_{DS}}{\partial I_D} = \frac{1}{\partial I_D / \partial V_{DS}} = \frac{1}{\lambda I_D}$$

$$g_m \propto \sqrt{\frac{I_D}{L}}$$

$$r_o \propto \frac{1}{\lambda I_D} \propto \frac{L}{I_D}$$

r_o has stronger dependence on L and I_D .

$$\text{Intrinsic Gain: } A_v = g_m r_o = \sqrt{2k'_n \frac{W}{L} I_D (1 + \lambda V_{DS})} \frac{1}{\lambda I_D} = \sqrt{\frac{2k'_n \frac{W}{L}}{I_D} \frac{1}{\lambda}}$$

Intrinsic gain A_v is inversely proportional to I_D

- can choose to include lambda in I_D or not, in the g_m expression

Q2. Use small-signal model to derive the equivalent resistance r_{eq} looking to the transistor terminal as shown in Fig. 2 (a) and Fig. 2 (b) below.

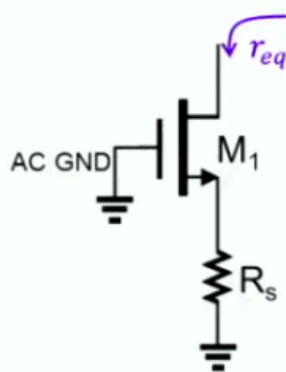


Fig. 2 (a)

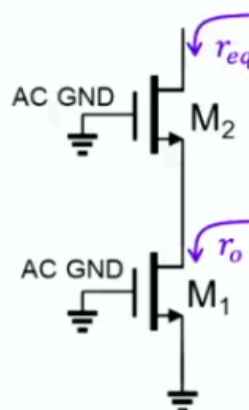
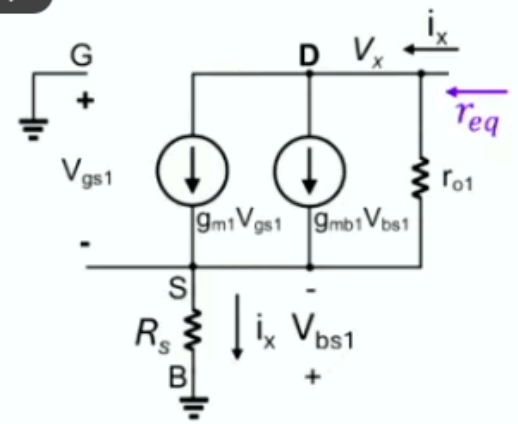
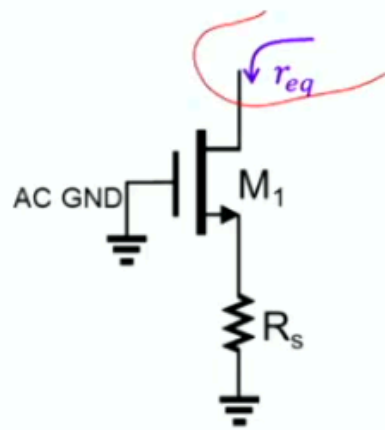


Fig. 2 (b)

(a)



$$i_x = g_{m1} v_{gs1} + g_{mb1} v_{bs1} + \frac{v_x + v_{bs1}}{r_{o1}}$$

Since $v_{bs1} = -i_x R_S = v_{gs1}$

$$i_x = -g_{m1} i_x R_S - g_{mb1} i_x R_S + \frac{v_x}{r_{o1}} - \frac{i_x R_S}{r_{o1}} \rightarrow i_x (1 + g_{m1} R_S + g_{mb1} R_S + \frac{R_S}{r_{o1}}) = \frac{v_x}{r_{o1}}$$

$$r_{eq} = \frac{v_x}{i_x} = (1 + g_{m1} R_S + g_{mb1} R_S + \frac{R_S}{r_{o1}}) r_{o1}$$

$$r_{eq} \approx [1 + (g_{m1} + g_{mb1}) R_S] r_{o1}, \quad (\text{if } [1 + (g_{m1} + g_{mb1}) R_S] r_{o1} \gg R_S)$$

- because R_S is smaller than r_{o1} , the $1 + \text{something}$ r_{o1} must be bigger than R_S , so can simplify
- can also think of the M_1 as a common gate, boosting R_S to bigger r_{eq} like in cascode

(b) Since the gate of M_1 is AC grounded, M_1 is simply a resistor r_{o1} . Replace r_{o1} with r_{o2} , R_S with r_{o1} and g_{m1} with g_{m2} in (a), we have

$$r_{eq} = \frac{v_x}{i_x}$$

$$= (1 + g_{m2} r_{o1} + g_{mb2} r_{o1} + \frac{r_{o1}}{r_{o2}}) r_{o2}$$

$$= (1 + g_{m2} r_{o1} + g_{mb2} r_{o1}) r_{o2} + r_{o1}$$

$$\approx [1 + (g_{m2} + g_{mb2}) r_{o1}] r_{o2}, \quad (\text{if } [1 + (g_{m2} + g_{mb2}) r_{o1}] r_{o2} \gg r_{o1})$$

Q3. Fig. 3 shows a common source amplifier with source degeneration resistor R_S . Derive the expression for amplifier voltage gain A_V .

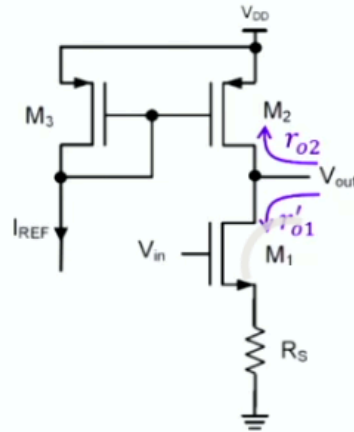
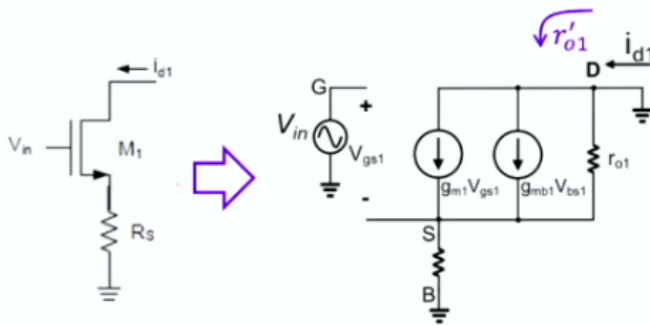


Fig. 3 CS amplifier with CSL and source degeneration resistor

Step 1: Find amplifier G_m , $G_m = i_{d1}/v_{in}$.



$$G_m = \frac{i_{d1}}{v_{in}}$$

$$i_{d1} = g_{m1}v_{gs1} + g_{mb1}v_{bs1} + \frac{v_{bs1}}{r_{o1}}$$

$$v_{bs1} = -i_{d1}R_z$$

$$v_{gs1} = v_{in} + v_{bs1} = v_{in} - i_{d1}R_z,$$

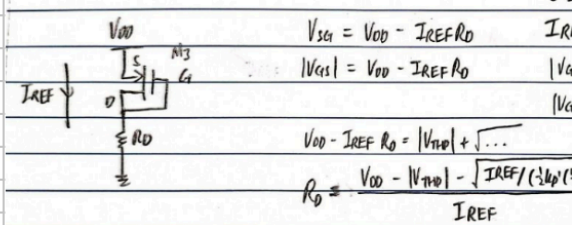
$$i_{d1} = g_{m1}(v_{in} - i_{d1}R_z) + g_{mb1}(-i_{d1}R_z) - \frac{i_{d1}R_z}{r_{o1}}$$

$$i_{d1}(1 + g_{m1}R_z + g_{mb1}R_z + \frac{R_z}{r_{o1}}) = g_{m1}v_{in}$$

$$G_m = \frac{i_{d1}}{v_{in}} = \frac{g_{m1}}{1 + g_{m1}R_z + g_{mb1}R_z + \frac{R_z}{r_{o1}}} = \frac{g_{m1}}{1 + (g_{m1} + g_{mb1} + \frac{1}{r_{o1}})R_z}$$

$$G_m \approx \frac{1}{(1 + \eta)R_z}, \quad \text{if } (g_{m1} + g_{mb1}) \gg \frac{1}{r_{o1}} \text{ and } (g_{m1} + g_{mb1})R_z \gg 1$$

- see the rough values for the simplification

choose ID1, uA	50
minimum (W/L)1 to hit gain target	221.1706503
choose (W/L)1	250
expected gm1, mS	2.121320344
expected ro1, kOhm	200
expected ro2, kOhm	200
2) Choose IREF, (W/L)3. Find (W/L)2, RD, VOD3 (= VOD2)	
	
choose IREF, uA	10
choose (W/L)3	300
RD, kOhms	132.6429774
(W/L)2	1500
VGS3 , V	0.473570226
VOD3, mV	23.57022604

- degeneration makes the GM of overall amplifier depends on the degeneration resistance instead of the gm of mosfet, which is not linear
- more linear gain $\Rightarrow$ improve linearity of amplifier

Step 2: Find output resistance of the amplifier using formula.

Looking into drain of M_1 :

$$r'_{o1} \approx [1 + (g_{m1} + g_{mb})R_s]r_{o1}$$

$$r_{out} = r_{o2} // r'_{o1}$$

$$\text{If } (g_{m1} + g_{mb})R_s \gg 1, \rightarrow r'_{o1} > r_{o2}$$

$$r_{out} = r_{o2} // r'_{o1} = r_{o2} // (g_{m1} + g_{mb})R_s r_{o1} \approx r_{o2}$$

- this assume that the 1 + something, the something is big enough to ignore r_{o1}

Step 3: Find voltage gain

$$A_v = \cancel{g_m} r_{out} = \cancel{r} \frac{g_{m1}}{1 + (g_{m1} + g_{mb1} + \frac{1}{r_{o1}})R_s} (r_{o2} // r'_{o1})$$

$$A_v \approx \frac{-r_{o2}}{(1 + \eta)R_s} \quad \text{If } (g_{m1} + g_{mb})R_s \gg 1$$

- with and without -ve is fine